

Appendix A. Proof of the conditional evaluator ordering

This appendix proves Theorem 1 of Section 4.3.1. Section A.1 fixes the setting and the conventions used throughout. Section A.2 states five lemmas and a corollary. Section A.3 carries out the induction over the reservation order, and Section A.4 gives the two counterexamples of Section 4.3.2 with complete timelines.

A.1 Setting and conventions

Fix a candidate wave W^k with its recorded sequence π^k (Equation (1)), and evaluate it under M_1 and M_2 from the common initial state of Section 3.1: every AMR, M_1 slot, and M_2 car is at floor f^0 with availability t_W , and no M_2 trip is stored. Both evaluations are deterministic. We assume $c \geq 2$, so that $Ec \geq E + 1$; the case $c = 1$ is treated in Section A.3.6.

Requests. The reservation of order o_j^k (Equations (18) to (22)) issues at most two elevator requests. The request q_j^S moves the assigned AMR to the source floor and exists when $G_{a_j}^{j-1} \neq s_{o_j^k}$; the request q_j^D delivers the order and exists when $s_{o_j^k} \neq d_{o_j^k}$. A request is identified by its label, (j, S) or (j, D) , and its origin and destination floors (g_q, h_q) :

$$(g_{q_j^S}, h_{q_j^S}) = (G_{a_j}^{j-1}, s_{o_j^k}), \quad (g_{q_j^D}, h_{q_j^D}) = (s_{o_j^k}, d_{o_j^k}). \quad (\text{A.1})$$

The request times, t_j^0 for q_j^S and t_j^P for q_j^D , depend on the evaluator and are written $t^{(1)}(q)$ and $t^{(2)}(q)$ below. Lemma A.2(i) shows that, under conditions (a) and (b), the two evaluations issue the same labelled requests with the same floors. The reservation order $<$ lists the requests of order o_1^k first, then those of o_2^k , and so on, with $q_j^S < q_j^D$. Both evaluators reserve requests in this order. As noted in Section 3.5.1, request times need not increase along $<$; the induction below follows $<$, exactly as the evaluators do.

Evaluator-specific quantities. As in Section 4.3.1, the evaluator is marked by the numeral 1 or 2: $t^{(1)}(q)$ and $t^{(2)}(q)$ are the request times, $D_1(q)$ and $D_2(q)$ the completion times returned by $\mathcal{R}_{M_1}$ and $\mathcal{R}_{M_2}$, and $G_{\text{sel}}^{(1)}(q)$ and $G_{\text{sel}}^{(2)}(q)$ the stored floors of the selected resources immediately before their state updates. Order completion times are t_{jkm}^C for $m \in \{M_1, M_2\}$, as in Equation (13). The letter i is reserved for ranks.

Resource states and ranks. Slot $v \in \mathcal{V}_1$ carries (B_v, G_v) , and car $e \in \mathcal{E}$ carries (B_e, G_e) and its latest trip record $((\bar{s}_e, \bar{d}_e), L_e, D_e, P_e)$, all as in Section 3.5.1. At the start of each reservation step, $x_{(1)} \leq \dots \leq x_{(Ec)}$ are the sorted slot availabilities $\{B_v\}$, and $y_{(1)} \leq \dots \leq y_{(E)}$ are the sorted car availabilities $\{B_e\}$, with the convention $y_{(E+1)} = +\infty$.

Two kinds of steps. Under M_1 , every request creates a new trip on the slot selected by Equation (23). Under M_2 , a request joins a stored trip when $\mathcal{B}(g, h, t) \neq \emptyset$ (Equations (25) and (26)) and otherwise creates a new trip (Equations (27) to (29)). A reservation step is a *new-trip step* when M_2 creates a new trip and a *joining step* when the request joins a stored M_2 trip; the two kinds are exhaustive. For a joining request on car e^* , the joined trip τ is the latest trip of e^* , because Equation (25) tests the latest record, and its completion satisfies $T_\tau = D_{e^*} = B_{e^*}$, because Equation (29) sets $B_{e^*} \leftarrow D_{e^*}$ when the trip is created and Equation (26) leaves B_{e^*} unchanged. Hence T_τ is one of the entries $\mathcal{Y}_{(1)}, \dots, \mathcal{Y}_{(E)}$.

Ties. Equations (18), (23), (26), and (27) resolve ties by index. The proof uses only two facts about the selections: a new trip is created on a resource of minimum availability, and a joined car's availability equals its latest trip's completion. The argument therefore holds for any tie-breaking rule; the index rules fix the realized paths on which conditions (b), (c), and (d) are checked.

Remark A.1 (server counts alone do not order completions). At every time, at most Ec requests are in transit under M_2 , because each of the E cars carries at most c requests; this equals the number of M_1 slots. The count does not by itself order completion times, for two reasons. First, each resource also stores a floor, and the earliest-availability rules (23) and (27) ignore it, so M_1 can select a distant slot while an M_2 car stands at the request's origin (Example A.2). Second, a joining request uses no additional car time, which can advance M_2 relative to M_1 (Example A.1). The induction below therefore tracks positions through condition (d) and joining through condition (c).

A.2 Lemmas

Lemma A.1 (per-trip parity). A request (g, h, t) that creates a new trip on a resource with availability B and stored floor G completes at

$$\theta(t, B, G; g, h) = \max\{t, B\} + \gamma|G - g| + \tau^L + \gamma|g - h| + \tau^U \quad (\text{A.2})$$

under both M_1 and M_2 . The function θ is nondecreasing in t , in B , and in $|G - g|$, and $\theta \geq B$.

Proof. Equation (24) is (A.2) for the selected slot. Equation (28) gives $D_{e^*} = L_{e^*} + \gamma|g - h| + \tau^U$ with $L_{e^*} = \max\{t, B_{e^*}\} + \gamma|G_{e^*} - g| + \tau^L$, which is (A.2) for the selected car. The number of requests on a trip enters neither formula. Monotonicity and $\theta \geq B$ are immediate. $\square$

Lemma A.2 (request alignment and monotone recursions). Suppose conditions (a) and (b) of Theorem 1 hold.

(i) The two evaluations generate the same requests in the same reservation order: for every j , the request q_j^S exists in both or in neither, and likewise q_j^D , with identical origin and destination floors and the same issuing AMR. Only request times and completion times may differ.

(ii) The request time of every request, and the completion time t_{jkm}^C of every order, are obtained from the completion times of requests reserved before them, and for t_{jkm}^C also from the order's own requests, by the same nondecreasing function for $m = M_1$ and $m = M_2$. For an order that issues no request, only earlier requests enter.

Proof. (i) Induct along π^k . By (a) the j -th order is the same in both evaluations, and by (b) so is its AMR a_j . Equation (22) sets the AMR's floor to the destination of the last order it served, or leaves it at f^0 , so by the induction hypothesis $G_{a_j}^{j-1}$ is the same in both evaluations. Whether q_j^S and q_j^D exist, and their floors, depend only on $G_{a_j}^{j-1}$, $s_{o_j^k}$, and $d_{o_j^k}$. The edge cases are covered: an order with $s_{o_j^k} = d_{o_j^k}$ issues no delivery request, an AMR already at the source issues no request q_j^S , and in every case the AMR's floor after the order is $d_{o_j^k}$. (ii) Equations (18) to (22) build t_j^0 , t_j^S , t_j^P , t_j^D , t_j^C , and $H_{a_j}^j$ from $t_W + r_{o_j^k}$, from $H_{a_j}^{j-1}$ (which is t_W or an earlier order's completion), and from the returned completion times, using only maxima with fixed numbers and additions of τ^P and τ^D . The AMR index is fixed by (b) and the branches of Equations (20) and (21) by part (i), so both evaluations apply the same composition of nondecreasing maps. Within an order, $q_j^S < q_j^D$, and the time of q_j^D uses the completion of q_j^S , which is reserved earlier. $\square$

Lemma A.3 (min-replacement identity). Let Z be a finite multiset of real numbers with sorted elements $\zeta_{(1)} \leq \dots \leq \zeta_{(|Z|)}$, set $\zeta_{(|Z|+1)} = +\infty$, and let $\omega \geq \zeta_{(1)}$. If Z' is obtained from Z by deleting one instance of $\zeta_{(1)}$ and inserting ω , then its sorted elements satisfy

$$\zeta'_{(i)} = \min\{\zeta_{(i+1)}, \max\{\zeta_{(i)}, \omega\}\}, \quad i = 1, \dots, |Z|. \quad (\text{A.3})$$

Proof. After the deletion, the sorted elements are $\zeta_{(2)} \leq \dots \leq \zeta_{(|Z|)}$. If $\omega > \zeta_{(i+1)}$, the i smallest elements of Z' are $\zeta_{(2)}, \dots, \zeta_{(i+1)}$, so $\zeta'_{(i)} = \zeta_{(i+1)}$, and the right side of (A.3) equals $\zeta_{(i+1)}$ because $\max\{\zeta_{(i)}, \omega\} = \omega > \zeta_{(i+1)}$. If $\omega \leq \zeta_{(i+1)}$, the i smallest elements are $\zeta_{(2)}, \dots, \zeta_{(i)}$ together with ω , so $\zeta'_{(i)} = \max\{\zeta_{(i)}, \omega\}$; for $i = 1$ this reads $\zeta'_{(1)} = \omega$, which uses $\omega \geq \zeta_{(1)}$. The right side equals the same value because $\max\{\zeta_{(i)}, \omega\} \leq \zeta_{(i+1)}$. $\square$

Lemma A.4 (ordering propagation). Let $x_{(1)} \leq \dots \leq x_{(Ec)}$ and $y_{(1)} \leq \dots \leq y_{(E)}$ satisfy $x_{(i)} \leq y_{(i)}$ for $i = 1, \dots, E$, with $Ec \geq E + 1$.

If one instance of $x_{(1)}$ is replaced by $\omega_1 \geq x_{(1)}$, one instance of $y_{(1)}$ by $\omega_2 \geq y_{(1)}$, and $\omega_1 \leq \omega_2$, then the new sorted values satisfy $x'_{(i)} \leq y'_{(i)}$ for $i = 1, \dots, E$.

Proof. By Lemma A.3, $x'_{(i)} = \min\{x_{(i+1)}, \max\{x_{(i)}, \omega_1\}\}$ and $y'_{(i)} = \min\{y_{(i+1)}, \max\{y_{(i)}, \omega_2\}\}$. For $i < E$, $x_{(i+1)} \leq y_{(i+1)}$ by hypothesis; for $i = E$, $x_{(E+1)}$ exists because $Ec \geq E + 1$, and $x_{(E+1)} \leq +\infty = y_{(E+1)}$. Also $\max\{x_{(i)}, \omega_1\} \leq \max\{y_{(i)}, \omega_2\}$. A minimum of smaller terms is smaller. $\square$

Lemma A.5 (recency of late completions). Throughout, the step is taken inside the coupled induction: conditions (a) and (b) are in force, and Lemma A.2 aligns the two evaluations' request sequences. Consider the start of the reservation step of a request q , and suppose $D_1(q') \leq D_2(q')$ for every request $q' < q$. For each rank $i = 1, \dots, E$, let $\mathcal{V}_1^>(y_{(i)}) = \{v \in \mathcal{V}_1 : B_v > y_{(i)}\}$ be the slots and $\mathcal{E}^>(y_{(i)}) = \{e \in \mathcal{E} : B_e > y_{(i)}\}$ the cars whose availability exceeds $y_{(i)}$. Then, for $i = 1, \dots, E$,

$$|\mathcal{V}_1^>(y_{(i)})| \leq \sum_{e \in \mathcal{E}^>(y_{(i)})} P_e \leq (E - i) c. \quad (\text{A.4})$$

Proof. Four observations.

1. Each car's availability B_e is nondecreasing over the evaluation. A new trip sets $B_e \leftarrow D_e \geq \max\{t, B_e\} + \tau^L + \tau^U > B_e$ (Equations (28) and (29)), and a joining request leaves B_e unchanged (Equation (26)). In particular $y_{(i)} \geq t_W$.
2. An M_2 trip that is no longer the latest trip of its car completed no later than the current $y_{(1)}$. When the next trip of that car was created, Equation (27) selected the car, so its availability at that moment, which equals the earlier trip's completion, was at most every car's availability at that moment; by observation 1, every availability has only grown since.
3. Hence every request $q' < q$ with $D_2(q') > y_{(i)}$ was reserved, as a creator or a joiner, on the latest trip of a car in $\mathcal{E}^>(y_{(i)})$, and $D_2(q')$ equals that car's current availability. The latest trip of car e carries $P_e \leq c$ requests, and $|\mathcal{E}^>(y_{(i)})| \leq E - i$ because at least i cars have availability at most $y_{(i)}$. So at most $\sum_{e \in \mathcal{E}^>(y_{(i)})} P_e \leq (E - i)c$ earlier requests have $D_2(q') > y_{(i)}$.
4. Each slot $v \in \mathcal{V}_1^>(y_{(i)})$ has $B_v > y_{(i)} \geq t_W$, so it has served at least one request, and $B_v = D_1(q'_v)$ for the last request q'_v it served (the update after Equation (24)). Distinct slots give

distinct requests. By the hypothesis, $D_2(q'_v) \geq D_1(q'_v) > y_{(i)}$. Observation 3 now bounds the number of such slots, which gives (A.4). $\square$

Corollary A.1 (a joining step preserves the ordering, any E). Assume conditions (a) and (b), and consider a step at which request q joins the stored trip τ of car e^* under M_2 . Suppose that $D_1(q') \leq D_2(q')$ for every $q' < q$, that $x_{(i)} \leq y_{(i)}$ for $i = 1, \dots, E$ at the start of the step, that $D_1(q) \geq x_{(1)}$, and that condition (c) holds for τ . Then $D_1(q) \leq D_2(q)$, and $x_{(i)} \leq y_{(i)}$ for $i = 1, \dots, E$ after the step.

Proof. After joining, τ contains at least two requests, so $q \in \mathcal{G}_\tau$, and Equation (44) gives $D_1(q) \leq T_\tau = D_2(q)$. Under M_2 , the availabilities are unchanged (Equation (26)), and $T_\tau = B_{e^*}$ (Section A.1). Joining requires $P_{e^*} \leq c - 1$ at the start of the step. Under M_1 , q creates a new trip on a slot with availability $x_{(1)}$, so one instance of $x_{(1)}$ is replaced by $\omega_1 = D_1(q)$, with $x_{(1)} \leq \omega_1 \leq T_\tau$. By Lemma A.3, $x'_{(i)} = \min\{x_{(i+1)}, \max\{x_{(i)}, \omega_1\}\}$. Two kinds of rank are possible.

- If $y_{(i)} \geq T_\tau$, then $x'_{(i)} \leq \max\{x_{(i)}, \omega_1\} \leq \max\{y_{(i)}, T_\tau\} = y_{(i)}$.
- If $y_{(i)} < T_\tau$, then $i \leq E - 1$ because $T_\tau \leq y_{(E)}$, and $e^* \in \mathcal{E}^>(y_{(i)})$. Lemma A.5, with $P_{e^*} \leq c - 1$, gives $|\mathcal{V}_1^>(y_{(i)})| \leq (E - i)c - 1 \leq Ec - i - 1$, where the last inequality is $i(c - 1) \geq 0$. At least $i + 1$ slots therefore have availability at most $y_{(i)}$, that is, $x_{(i+1)} \leq y_{(i)}$, and $x'_{(i)} \leq x_{(i+1)} \leq y_{(i)}$.

Since the y values are unchanged, all E coordinates of the ordering hold after the step. $\square$

The corollary uses condition (c) only for the joining request. The request that created τ was handled when τ was created, in a new-trip step.

A.3 Proof of Theorem 1

A.3.1 Invariants

The proof proceeds by induction along the reservation order $<$. After the step of each request, (I1) and (I2) hold for every request q' reserved so far, and (I3) holds for the current resource states.

- (I1) Request-time ordering: $t^{(1)}(q') \leq t^{(2)}(q')$.
- (I2) Completion-time ordering: $D_1(q') \leq D_2(q')$.
- (I3) Availability ordering of the E earliest resources: $x_{(i)} \leq y_{(i)}$ for $i = 1, \dots, E$.

(I1) follows from (I2): by Lemma A.2(ii), the request time of q is the same nondecreasing function of the completions of requests $q' < q$ in both evaluations, so (I2) for those requests gives $t^{(1)}(q) \leq t^{(2)}(q)$.

A.3.2 Base case

Before the first request, every slot and every car has availability t_W and floor f^0 , so (I3) holds with equality, and (I1) and (I2) are vacuous. No elevator completion enters the time of the first request, so, by Lemma A.2(ii), its request time is the same in both evaluations. M_2 stores no trip, so the request creates a new trip in both evaluations, on resources with availability t_W at floor f^0 . By Lemma A.1, its two completion times coincide. (I3) is restored by Lemma A.4 with $\omega_1 = \omega_2$.

A.3.3 Inductive step: new-trip step

Both evaluations create a new trip. Under M_1 , the selected slot has availability $x_{(1)}$ and stored floor $G_{\text{sel}}^{(1)}(\varrho)$; under M_2 , the selected car has availability $y_{(1)}$ and stored floor $G_{\text{sel}}^{(2)}(\varrho)$. By (I1), $t^{(1)}(\varrho) \leq t^{(2)}(\varrho)$; by (I3) at $i = 1$, $x_{(1)} \leq y_{(1)}$; by condition (d), $|G_{\text{sel}}^{(1)}(\varrho) - g_\varrho| \leq |G_{\text{sel}}^{(2)}(\varrho) - g_\varrho|$. Lemma A.1 then gives

$$D_1(\varrho) = \theta(t^{(1)}(\varrho), x_{(1)}, G_{\text{sel}}^{(1)}(\varrho); g_\varrho, h_\varrho) \leq \theta(t^{(2)}(\varrho), y_{(1)}, G_{\text{sel}}^{(2)}(\varrho); g_\varrho, h_\varrho) = D_2(\varrho), \quad (\text{A.5})$$

which extends (I2). M_1 replaces one instance of $x_{(1)}$ by $\omega_1 = D_1(\varrho)$, and M_2 replaces one instance of $y_{(1)}$ by $\omega_2 = D_2(\varrho)$. Lemma A.1 gives $\omega_1 \geq x_{(1)}$ and $\omega_2 \geq y_{(1)}$, and (A.5) gives $\omega_1 \leq \omega_2$, so Lemma A.4 restores (I3).

A.3.4 Inductive step: joining step

Under M_1 , ϱ creates a new trip on a slot with availability $x_{(1)}$, so $D_1(\varrho) \geq x_{(1)}$ by Lemma A.1. The induction hypothesis supplies (I2) for all $\varrho' < \varrho$ and (I3) at the start of the step, and condition (c) holds for the joined trip. Corollary A.1 gives $D_1(\varrho) \leq D_2(\varrho)$, which extends (I2), and (I3) after the step.

A.3.5 Conclusion

By induction, $D_1(\varrho) \leq D_2(\varrho)$ for every request. By Lemma A.2(ii), each order completion t_{jkm}^C is the same nondecreasing function of the completions of the order's own and earlier requests for $m = M_1$ and $m = M_2$; this covers orders that issue no request and only inherit earlier delays through the AMR's availability. Hence $t_{jkm_1}^C \leq t_{jkm_2}^C$ for every j , and Equation (13) gives

$$C_{\max}(W^k, \pi^k; M_1) = \max_{j \in \mathcal{J}_n} t_{jkm_1}^C - t_W \leq \max_{j \in \mathcal{J}_n} t_{jkm_2}^C - t_W = C_{\max}(W^k, \pi^k; M_2),$$

which, with the request-level ordering, is Equation (46). $\square$

A.3.6 The case $c = 1$

With $c = 1$, no request can join a trip: initially no trip is stored, and afterwards every stored trip has $P_e = 1$, so $\mathcal{B}(g, h, t) = \emptyset$ in Equation (25). Every M_2 request therefore creates a new trip, and M_2 consists of E single-request resources, like the $E \cdot 1$ slots of M_1 . Both evaluators start from identical resource states, select resources by the same rule, and update them by the same formula (A.2). By induction along the reservation order, identical completions give identical AMR states and assignments (Equations (18) to (22)), hence identical requests and request times, so condition (b) holds automatically. The two evaluations therefore coincide, and Equation (46) holds with equality. Because the initial resources are identical, the equality holds for any label order used in both evaluators, not only for the index order of Section 3.5.

A.3.7 What the proof uses

Conditions (a) and (b) enter through Lemma A.2, which aligns the requests and orders their times. Condition (d) is used only in new-trip steps, and condition (c) only in joining steps, for the joining request. Everything else is derived from the reservation rules of Section 3.5. Read in the contrapositive, the theorem classifies failures: when (a) and (b) hold, a reversal of either inequality in Equation (46) requires condition (c) or condition (d) to fail on the paired reservation paths. The converse does not hold, since a violated condition need not produce a reversal (Section 4.3.2).

A.4 Neither (c) nor (d) can be dropped: two examples

Both examples use $t_W = 0$, $f^0 = 1$, $E = 1$, $c = 2$, $\gamma = 5$, $\tau^L = \tau^U = 2$, and $\tau^P = \tau^D = 5$, with times in seconds, as in Section 4.3.2. They are hand-constructed minimal instances, not drawn from any experiment. In each, conditions (a) and (b) hold, exactly one of (c) and (d) fails, and the makespan ordering reverses, so neither condition can be dropped from Theorem 1. A violated condition alone need not produce a reversal (Section 4.3.2). Each example can be reproduced by one deterministic call of the closed-form evaluators.

Example A.1 (shared-trip overtaking: condition (c) cannot be dropped)

Let $F = 3$ and $A = 2$. Order o_1 travels from floor 1 to floor 3 with $r_{o_1} = 0$, and order o_2 from floor 1 to floor 3 with $r_{o_2} = 2$; the recorded sequence (Equation (1)) is $\pi^k = (o_1, o_2)$. Order o_1 goes to AMR 1 (both AMRs are available at 0; index rule). The AMR is at the source floor, so there is no request q_1^S ; the delivery request q_1^D , with floors (1, 3), is issued at $t_1^P = 5$ in both evaluations. Order o_2 goes to AMR 2, whose availability 0 is below AMR 1's availability 24 in both evaluations, so (b) holds; $t_2^0 = 2$, and q_2^D , with floors (1, 3), is issued at $t_2^P = 7$ in both evaluations.

Request	Evaluator	Resource and action	Wait until	Reposition	Load	Travel	Unload	Completion	Order finishes
q_1^D : 1 to 3, issued at 5	M_1	slot 1 ($B = 0$, floor 1), new trip	5	none	5 to 7	7 to 17	17 to 19	19	24
q_2^D : 1 to 3, issued at 7	M_1	slot 2 ($B = 0$, floor 1), new trip	7	none	7 to 9	9 to 19	19 to 21	21	26
q_1^D : 1 to 3, issued at 5	M_2	car 1 ($B = 0$, floor 1), new trip, $L = 7$	5	none	5 to 7	7 to 17	17 to 19	19	24
q_2^D : 1 to 3, issued at 7	M_2	joins car 1's trip (pair (1,3), $P = 1 < 2$, $7 \leq L = 7$)						19	24

The makespans are $C_{\max}(W^k, \pi^k; M_1) = 26$ and $C_{\max}(W^k, \pi^k; M_2) = 24$. Condition (d) holds: the only request with a new trip in both evaluations is q_1^D , with $|1 - 1| \leq |1 - 1|$. Condition (c) fails: the trip of car 1 has $\mathcal{G}_\tau = \{q_1^D, q_2^D\}$ and $T_\tau = 19$, but $D_1(q_2^D) = 21 > 19$. The shared trip began loading at time 5, when q_1^D was issued (the car had been free since time 0), and q_2^D joined at its request time 7, as the loading window closed ($7 \leq L = 7$). The private M_1 slot cannot begin loading before the request time 7. The two-second reversal equals the ready-time stagger.

Example A.2 (adverse repositioning: condition (d) cannot be dropped)

Let $F = 8$ and $A = 1$. Order o_1 travels from floor 1 to floor 8 and order o_2 from floor 8 to floor 7, both with ready-time offset 0; the recorded sequence (Equation (1)) is $\pi^k = (o_1, o_2)$. With one AMR, (b) holds trivially. For o_1 , the delivery request q_1^D , with floors (1, 8), is issued at 5, and the order finishes at $44 + 5 = 49$ in both evaluations, leaving the AMR at floor 8. For o_2 , the AMR is already at the source floor 8, so there is no request q_2^S ; $t_2^0 = 49$, and q_2^D , with floors (8, 7), is issued at $t_2^P = 54$ in both evaluations.

Request	Evaluator	Resource and action	Wait until	Reposition	Load	Travel	Unload	Completion	Order finishes
q_1^D : 1 to 8, issued at 5	M_1	slot 1 ($B = 0$, floor 1), new trip	5	none	5 to 7	7 to 42	42 to 44	44	49
q_2^D : 8 to 7, issued at 54	M_1	slot 2 ($B = 0 < 44$, floor 1), new trip	54	54 to 89 (floor 1 to 8)	89 to 91	91 to 96	96 to 98	98	103
q_1^D : 1 to 8, issued at 5	M_2	car 1 ($B = 0$, floor 1), new trip	5	none	5 to 7	7 to 42	42 to 44	44	49
q_2^D : 8 to 7, issued at 54	M_2	car 1 ($B = 44$, floor 8), new trip; no match with stored pair (1,8)	54	none	54 to 56	56 to 61	61 to 63	63	68

The makespans are $C_{\max}(W^k, \pi^k; M_1) = 103$ and $C_{\max}(W^k, \pi^k; M_2) = 68$. Condition (c) holds vacuously, since no M_2 trip carries two requests. Condition (d) fails at q_2^D , where both evaluations create a new trip: $|G_{\text{sel}}^{(1)}(q_2^D) - g_{q_2^D}| = |1 - 8| = 7 > 0 = |G_{\text{sel}}^{(2)}(q_2^D) - g_{q_2^D}|$. The 35-second reversal is the seven-floor repositioning leg: the earliest-availability rule (23) selects slot 2, idle since time 0 at floor 1, while slot 1 and the M_2 car stand at floor 8. The example has no shared trip, so shared-trip overtaking is not the only channel of reversal.